

Skill Test 1: Cloud and Containers

Test Name - Cloud and Containers

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Batch - 16A

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Skill Test Github Link - <https://github.com/saleeminfodevops/Microservices-Task.git>

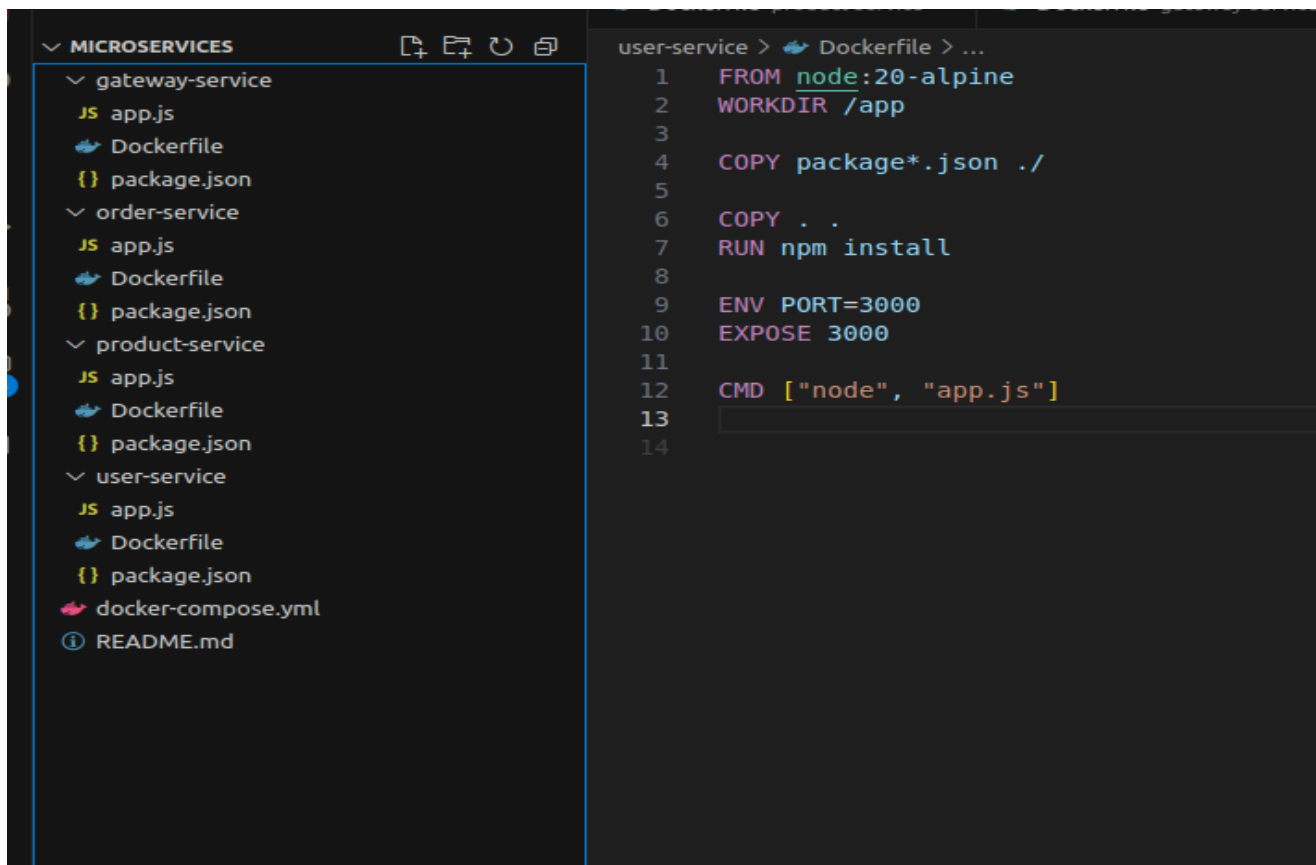
1) Installed Docker and created a network:

```
docker network create ms-net
```

2) Forked the repo to my github

<https://github.com/saleeminfodevops/Microservices-Task.git>

3) Created Dockerfile in each application folder:



4) Created Docker-compose.yml in the main folder:

Microservices-Task/Microservices/docker-compose.yml

```
1 services:
2   > Run All Services
3   > Run Service
4   users:
5     container_name: user-service
6     build:
7       context: ./user-service
8       dockerfile: Dockerfile
9     ports:
10      - 3000:3000
11     networks:
12      - ms-net
13   > Run Service
14   product:
15     container_name: product-service
16     build:
17       context: ./product-service
18       dockerfile: Dockerfile
19     ports:
20      - 3001:3001
21     networks:
22      - ms-net
23   > Run Service
24   order:
25     container_name: order-service
26     build:
27       context: ./order-service
28     dockerfile: Dockerfile
```

5) Started the Docker compose:

docker compose up -- build

```
=> CACHED [users 5/5] RUN npm install 0.0s
=> CACHED [product 3/5] COPY package*.json ./ 0.0s
=> CACHED [product 4/5] COPY . . 0.0s
=> CACHED [product 5/5] RUN npm install 0.0s
=> CACHED [gateway 3/5] COPY package*.json ./ 0.0s
=> CACHED [gateway 4/5] COPY . . 0.0s
=> CACHED [gateway 5/5] RUN npm install 0.0s
=> CACHED [order 3/5] COPY package*.json ./ 0.0s
=> CACHED [order 4/5] COPY . . 0.0s
=> CACHED [order 5/5] RUN npm install 0.0s
=> [users] exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:b7b92704c028a9b1d5c7372156a6da09fcfe54fd69a9e11ab621402a901caab8 0.0s
=> => naming to docker.io/library/microservices-users 0.0s
=> [product] exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:2c8175313dc9d7e84a7e772f39cb9219019de3235dbb51edf622ee04108bc10e 0.0s
=> => naming to docker.io/library/microservices-product 0.0s
=> [gateway] exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:ab6d949fd037557485614e62586f64elbc61bcf4592e4d5d4b8183a836b553ce 0.0s
=> => naming to docker.io/library/microservices-gateway 0.0s
=> [order] exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:fd4578ed2d1a0306ffdf3f794b5ee3ccf7b789f87b2926d1b4a83d89118ce8e8 0.0s
=> => naming to docker.io/library/microservices-order 0.0s
=> [users] resolving provenance for metadata file 0.0s
=> [gateway] resolving provenance for metadata file 0.0s
=> [product] resolving provenance for metadata file 0.0s
=> [order] resolving provenance for metadata file 0.0s
[+] up 4/4
✓ Image microservices-order Built 6.9s
✓ Image microservices-gateway Built 6.9s
✓ Image microservices-users Built 6.9s
✓ Image microservices-product Built 6.9s
Attaching to gateway-service, order-service, product-service, user-service
order-service | Order service running on port 3002
product-service | Product service running on port 3001
gateway-service | Gateway service running on port 3003
user-service | User service running on port 3000

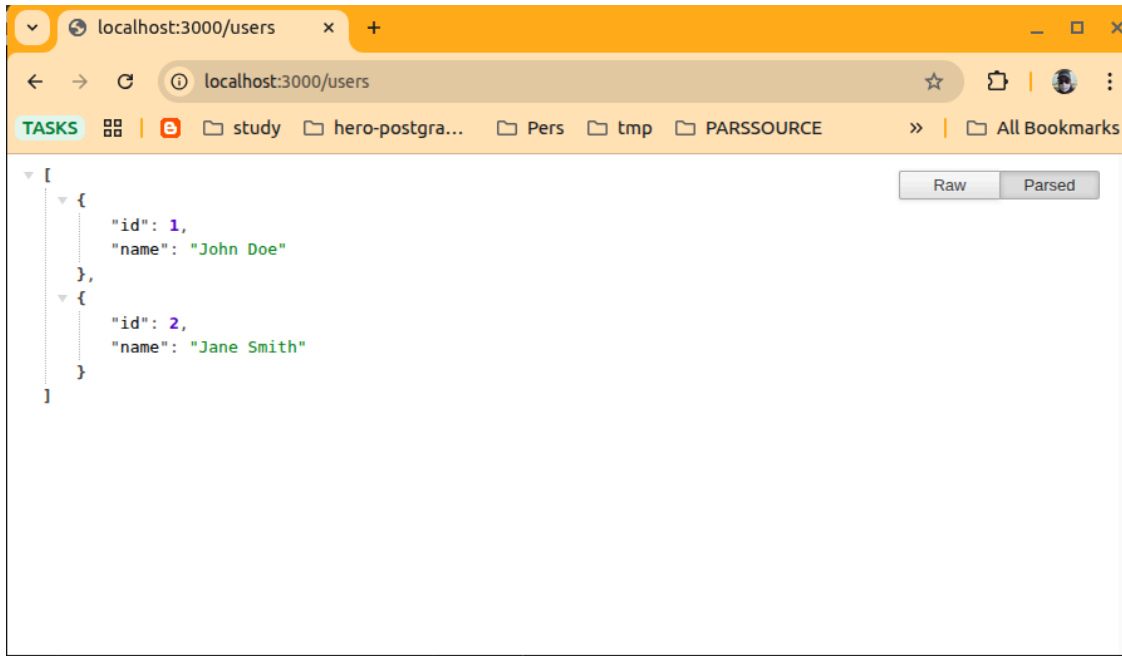
```

```
o testing@inblrit-002:~/study/skiltest/Microservices-Task/Microservices$ docker compose up --build
[+] Building 6.9s (37/37) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 2.208B 0.0s
=> [order internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 173B 0.0s
=> [gateway internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 173B 0.0s
=> [users internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 173B 0.0s
=> [product internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 173B 0.0s
=> [order internal] load metadata for docker.io/library/node:20-alpine 6.5s
[auth] library/node:pull token for registry-1.docker.io 0.0s
=> [users internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [gateway internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [product internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [order internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [product 1/5] FROM docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293 0.0s
=> [users internal] load build context 0.0s
=> => transferring context: 90B 0.0s
=> [order internal] load build context 0.0s
=> => transferring context: 90B 0.0s
=> [gateway internal] load build context 0.0s
=> => transferring context: 90B 0.0s
=> [product internal] load build context 0.0s
=> => transferring context: 90B 0.0s
=> CACHED [order 2/5] WORKDIR /app 0.0s
=> CACHED [users 3/5] COPY package*.json ./ 0.0s
=> CACHED [users 4/5] COPY . . 0.0s
=> CACHED [users 5/5] RUN npm install 0.0s
=> CACHED [product 3/5] COPY package*.json ./ 0.0s
=> CACHED [product 4/5] COPY . . 0.0s
=> CACHED [product 5/5] RUN npm install 0.0s
=> CACHED [gateway 3/5] COPY package*.json ./ 0.0s
=> CACHED [gateway 4/5] COPY . . 0.0s
=> CACHED [gateway 5/5] RUN npm install 0.0s
=> CACHED [order 3/5] COPY package*.json ./ 0.0s
=> CACHED [order 4/5] COPY . . 0.0s
=> CACHED [order 5/5] RUN npm install 0.0s
=> [users] exporting to image 0.0s

```

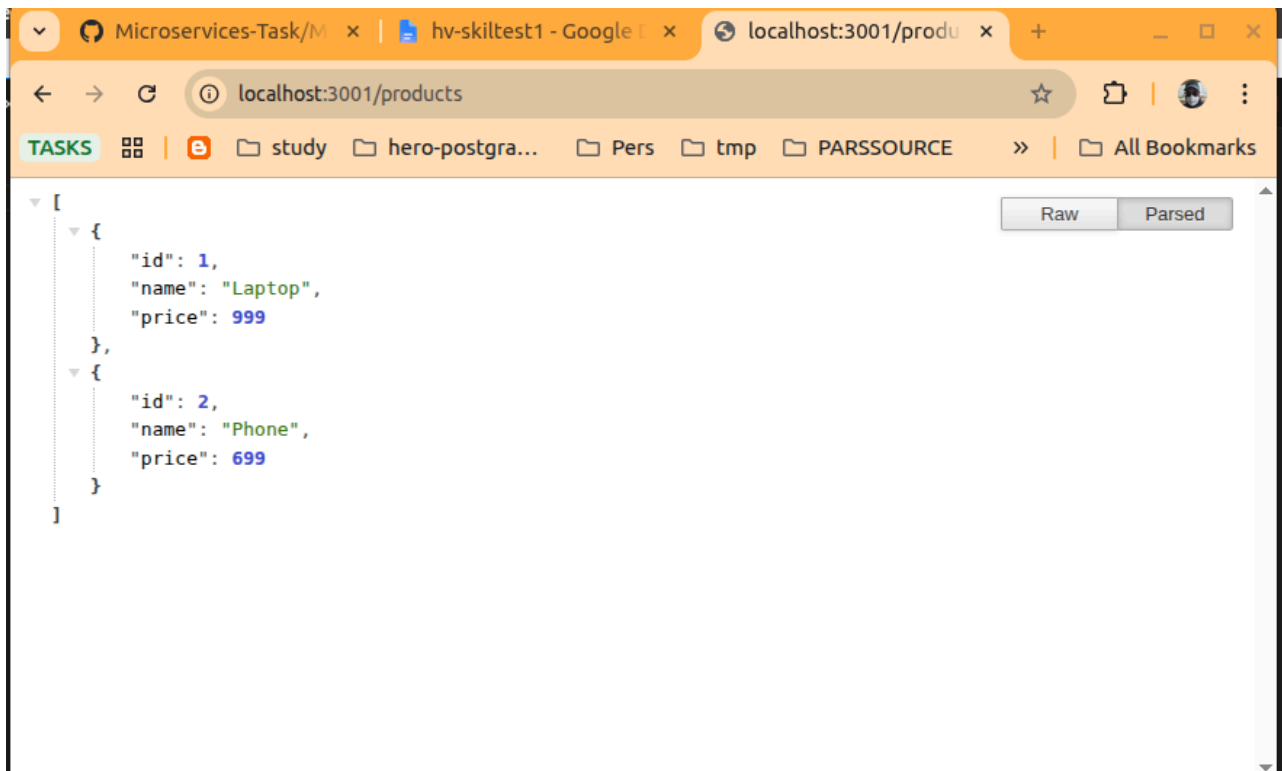
6) Verified by accessing from the Browser:

User-service:



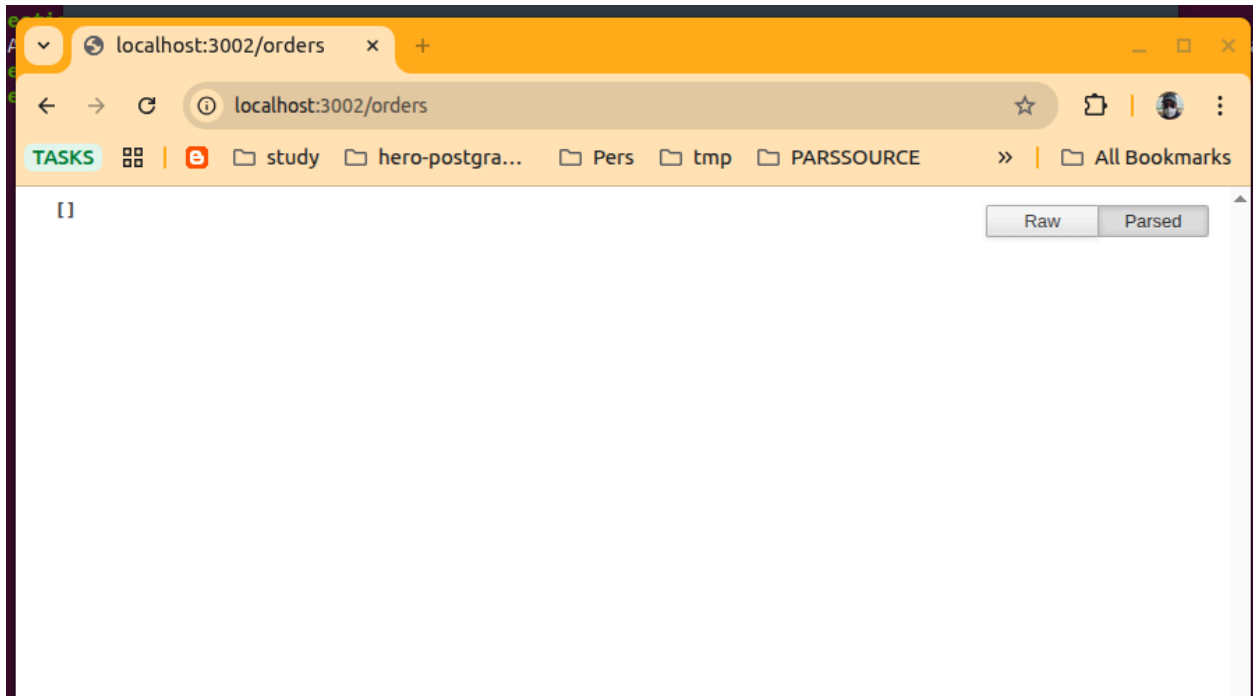
```
[{"id": 1, "name": "John Doe"}, {"id": 2, "name": "Jane Smith"}]
```

Products-service:

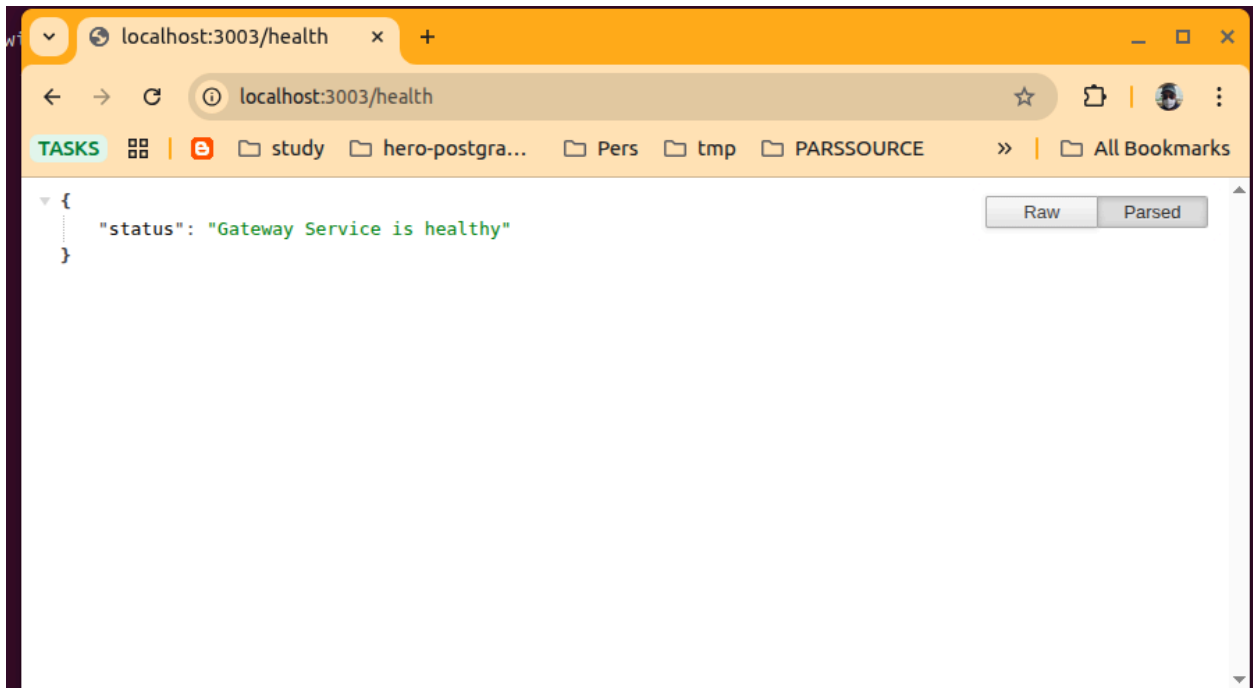


```
[{"id": 1, "name": "Laptop", "price": 999}, {"id": 2, "name": "Phone", "price": 699}]
```

Orders-Service:



Gateway-Service:



Gateway using the API:

